Experiment - 4

Study and Demonstration of Data Communication Using OSI and TCP/IP Models in Cisco Packet Tracer

Aim

To understand and demonstrate how data travels through the **OSI model and TCP/IP model** during communication between two computers using Cisco Packet Tracer.

Objectives

- To create a simple computer network using Cisco Packet Tracer.
- To configure IP addresses for two PCs.
- To establish communication between the PCs through a switch.
- To verify connectivity using the ping command.
- To observe packet movement using Simulation Mode.
- To understand the functions of the seven OSI layers.
- To relate the OSI model to the TCP/IP model.
- To observe IP, MAC, and protocol information in transmitted packets.

Software and Hardware Requirements

- Cisco Packet Tracer
- Two PCs
- One Cisco 2960 switch
- Copper straight-through cables

Scenario

A small computer network is created to demonstrate how data is transmitted from one computer to another.

The network consists of **PC0, a 2960 switch, and PC1**. PC0 and PC1 are connected to the switch using copper straight-through cables. Both PCs are assigned IP addresses from the same network.

Network structure:

PC0 → Switch → PC1

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

No default gateway is required because both PCs are in the same local network.

Network Configuration

PC0 is configured with:

- IP Address: **192.168.1.10**
- Subnet Mask: **255.255.255.0**

PC1 is configured with:

- IP Address: **192.168.1.20**
- Subnet Mask: **255.255.255.0**

The PCs are connected to the switch using:

- PC0 FastEthernet0 → Switch FastEthernet0/1
- PC1 FastEthernet0 → Switch FastEthernet0/2

Procedure

1. Open Cisco Packet Tracer.
2. Place two PCs in the workspace.
3. Add a Cisco 2960 switch.
4. Connect both PCs to the switch using copper straight-through cables.
5. Configure the IP address of PC0 as 192.168.1.10.
6. Configure the IP address of PC1 as 192.168.1.20.
7. Open the Command Prompt on PC0.
8. Enter ping 192.168.1.20.
9. Verify that PC1 sends successful replies.
10. Change Packet Tracer from **Realtime Mode to Simulation Mode**.
11. Use **Add Simple PDU** to create a packet from PC0 to PC1.

12. Use **Capture/Forward** to observe the packet moving through the network.
13. Open the packet details to observe IP addresses, MAC addresses, and protocol information.

Working Principle

- When PC0 sends data to PC1, the data passes through different layers of the networking model.
- At the **Application Layer**, the communication begins with an application-generated message.
- The **Transport Layer** manages the delivery of data using protocols such as TCP or UDP.
- The **Network Layer** uses IP addresses to identify the source and destination devices.
- The **Data Link Layer** uses MAC addresses and creates frames for communication across the local network.
- The **Physical Layer** transmits the information as electrical or physical signals through the network cable.
- The switch receives the frame and forwards it toward PC1. At PC1, the information is processed in the reverse direction until the original data reaches the application.

OSI Model Analysis

The OSI (Open Systems Interconnection) Model consists of seven layers. Each layer performs a specific function in the communication process between two devices in a network.

Layer 7 – Application Layer:
The Application layer provides network services directly to applications used by the user. It enables applications to communicate over the network.

Layer 6 – Presentation Layer:
The Presentation layer handles the format and representation of data. It is responsible for data translation, encryption, decryption, and compression when required.

Layer 5 – Session Layer:
The Session layer establishes, manages, and terminates communication sessions between devices. It helps maintain an active connection during communication.

Layer 4 – Transport Layer:
The Transport layer provides reliable data delivery between devices. It uses protocols such as TCP and UDP to control data transmission.

Layer 3 – Network Layer:
The Network layer is responsible for logical addressing and routing. It uses IP addresses to determine the destination and route data between networks.

Layer 2 – Data Link Layer:
The Data Link layer provides communication between devices on the same network. It uses MAC addresses and organizes data into frames.

Layer 1 – Physical Layer:
The Physical layer is the lowest layer of the OSI model. It transfers data in the form of bits through physical media such as network cables.

Memory Trick:

A P S T N D P – *All People Seem To Need Data Processing*

TCP/IP Model Analysis

The TCP/IP model can be represented using four layers:

TCP/IP Layer	Example/Function
Application	HTTP, DNS, FTP
Transport	TCP, UDP
Internet	IP
Network Access	Ethernet, Wi-Fi

The relationship between the two models is:

- OSI Application + Presentation + Session → **TCP/IP Application**
- OSI Transport → **TCP/IP Transport**

- OSI Network → **TCP/IP Internet**
- OSI Data Link + Physical → **TCP/IP Network Access**

Packet Observation

Using Simulation Mode, the packet can be inspected as it travels from PC0 to PC1.

The following information can be observed:

- Source IP address
- Destination IP address
- Source MAC address
- Destination MAC address
- Protocol information
- Packet movement between devices

This demonstrates that different layers are responsible for adding and processing different types of communication information.

Real-Life Example

Consider sending the message **"Hello"** from one computer to another.

The message starts at the application level and moves down through the networking layers. Information such as transport details, IP addressing, and MAC addressing is handled by the appropriate layers. Finally, the data is converted into signals and transmitted through the network.

At the receiving computer, the process occurs in reverse order, and the original message is delivered to the application.

Expected Result

The two PCs should successfully communicate with each other. The ping command should produce replies from PC1, confirming network connectivity.

In Simulation Mode, the packet should successfully travel through:

PC0 → Switch → PC1

The packet details should demonstrate the role of IP addresses, MAC addresses, and networking protocols.

Conclusion

This case study demonstrates how data communication takes place between two computers using Cisco Packet Tracer. The experiment helps in understanding the functions of the **OSI and TCP/IP models** and how different layers participate in network communication. Simulation Mode provides a visual representation of packet movement, making the concepts of IP addressing, MAC addressing, frames, and data transmission easier to understand.